Advanced NLP – Exercise 1

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GitHub repository: https://github.com/DanAbergel/ANLP_EX1

1 Open Questions

Question 1 – Three QA datasets that annotate intrinsic concepts

A QA dataset is *intrinsic* when the question–answer format is used as a tool to annotate a property of language itself (e.g., predicate–argument structure, discourse relations, coreference), rather than as the end task of answering questions about the world. In an intrinsic QA dataset, the answer can always be recovered from the given sentence or paragraph alone, with no reliance on external world knowledge. We list three such datasets below.

1. **QA-SRL** he2015question – *Semantic Role Labeling as Question Answering.* QA-SRL annotates predicate–argument structure by asking natural-language wh-questions about each verb in a sentence and marking the answer span. For example, given the sentence *"The pilot landed the plane safely after the engine failed,"* the predicate *landed* is annotated with questions such as *"Who landed something?"* (*the pilot*), *"What did someone land?"* (*the plane*), and *"When did someone land something?"* (*after the engine failed*). **Why intrinsic:** the annotation targets the predicate–argument structure of the sentence – a core linguistic property – and every answer is a span within the sentence itself, requiring no world knowledge.

2. **QAMR** michael2018crowdsourcing – *Question–Answer Meaning Representations.* QAMR represents the propositional meaning of a sentence as a set of free-form question–answer pairs covering predicate–argument relations and modifiers. For example, given *"After the storm, the mayor ordered an evacuation of the coastal areas,"* the annotation includes pairs such as *"Who ordered something?"* (*the mayor*), *"What was evacuated?"* (*the coastal areas*), and *"What kind of areas?"* (*coastal*). **Why intrinsic:** the dataset captures the meaning representation of a single sentence; the questions probe its internal semantic structure rather than any external fact, so success depends purely on linguistic understanding.

3. **QADiscourse** pyatkin2020qadiscourse – *Discourse Relations as Question Answering.* QADiscourse annotates inter-clausal discourse relations (cause, result, contrast, condition, etc.) by asking natural-language questions whose answer reveals the relation between two clauses. For example, given *"The bridge was closed for repairs. Commuters had to take a long detour,"* the annotation includes *"Why did commuters have to take a detour?"* (*because the bridge was closed for repairs* – a *cause* relation). **Why intrinsic:** the target is the discourse structure of the text – how clauses connect logically – which is an intrinsic property of language organization, independent of the topic or content of the passage.

These three datasets cover three distinct levels of linguistic structure – predicate-level (QA-SRL), sentence-level (QAMR), and discourse-level (QADiscourse) – showing how the QA format can be repurposed as an annotation device for intrinsic linguistic properties rather than as an end-task over world knowledge.

Question 2 – Inference-time scaling methods

Inference-time scaling refers to spending additional compute at test time – rather than at training time – to obtain better answers from a pretrained language model. The course covered three families of inference-time scaling methods, discussed in turn below.

(a) Description of each method

1. Self-Consistency wang2022selfconsistency.

- **Description.** Instead of greedy-decoding a single chain-of-thought, sample N diverse reasoning chains for the same prompt (with non-zero temperature), then take a *majority vote* over their final answers. The most frequent final answer is returned.
- **Advantages.** Simple to implement (no extra model, no extra training); robust and well-documented improvements on math and science reasoning benchmarks; complements any CoT-prompted model.
- **Computational bottlenecks.** Multiplies decoding cost by N (each chain is a full autoregressive generation). The bottleneck is GPU compute and KV-cache memory across the N parallel chains.
- **Parallelizable? Yes – embarrassingly parallel.** The N samples are independent and can be batched in a single forward pass on the same GPU, which exploits batched inference efficiently.

2. Verifier-based Best-of-N cobbe2021training,lightman2023letsverify.

- **Description.** Sample N candidate solutions from the generator model, then score each one with a separate trained *verifier*. Two flavours were discussed: *outcome reward models* (ORM) that score only the final answer, and *process reward models* (PRM) that score each intermediate reasoning step. The candidate with the highest verifier score is selected.
- **Advantages.** Often outperforms self-consistency because the verifier can identify a correct minority answer; PRMs give finer-grained feedback than majority voting; verifiers can also be used at training time (e.g., to filter CoT data).
- **Computational bottlenecks.** $N\times$ generator decoding *plus* N (or more, for PRMs) verifier forward passes. The verifier itself must fit in GPU memory alongside the generator – additional parameter footprint.
- **Parallelizable? Yes.** Generation of the N candidates is independent and batchable; verifier scoring is also independent across candidates. End-to-end parallelism is comparable to self-consistency, modulo the extra verifier cost.

3. Long Chain-of-Thought reasoning (O1 / DeepSeek-R1-style) openai2024o1,deepseek2025r1.

- **Description.** Rather than running many short chains, the model is trained (typically via reinforcement learning with a verifiable reward) to produce a *single but very long* chain-of-thought, internally performing **planning**, **backtracking** (“wait, that doesn’t work, let me try again”), and **self-evaluation**. Test-time compute is scaled by allowing the model to “think” for many more tokens within one trajectory. Empirically, the average response length grows substantially with RL training (the “aha moment” observed in DeepSeek-R1-Zero).

- **Advantages.** State-of-the-art performance on hard reasoning benchmarks (AIME, scientific QA, competition math); the model itself learns when to backtrack or re-evaluate, so no external verifier or vote is required at inference time; a single coherent reasoning trace is easier to inspect and debug than N separate chains.
- **Computational bottlenecks.** Each generation is intrinsically *sequential*: the KV-cache grows linearly with the number of reasoning tokens, and decoding latency scales with the (much longer) sequence length. The bottleneck is therefore memory for the KV cache and wall-clock latency, not aggregate FLOPs across samples.
- **Parallelizable? No – not within a single trajectory.** Token generation is autoregressive and cannot be parallelised along the sequence dimension. Different prompts can of course be batched, but the cost of any single prediction grows with thinking length.

(b) Which method on a single large-memory GPU for a complex scientific reasoning task?

On a *single GPU with large memory capacity* for a complex scientific reasoning task, I would choose **long chain-of-thought reasoning with an O1 / DeepSeek-R1-style model**, for the following reasons:

1. **Memory is the binding constraint, and we have it.** A long reasoning trace requires a large KV cache, which scales linearly with thinking length. A single large-memory GPU is the ideal hardware profile for this: we do not need multi-GPU sharding, but we do need enough memory to hold the entire growing context. This is exactly what the scenario provides.
2. **Self-consistency and Best-of-N scale across samples, not depth.** Their gains come from parallelism over N independent rollouts, which benefits more from many GPUs or large batch throughput than from per-GPU memory. On a *single* GPU, the value of extra memory is mostly wasted under these methods.
3. **Empirical evidence.** On hard scientific and mathematical reasoning benchmarks (AIME, science olympiads), reasoning models that internalise planning, backtracking, and self-evaluation currently outperform sampling-based scaling approaches at comparable compute budgets.
4. **No external verifier required.** The single-model setup avoids having to fit a generator *and* a separate verifier in memory simultaneously, which simplifies deployment on one GPU.

A reasonable alternative would be self-consistency, which is also a strong fit for a single GPU thanks to batched parallel sampling; but for tasks where the difficulty lies in long, structured reasoning (rather than in noise across short chains), long CoT reasoning is the more compute-efficient choice on this hardware profile.

2 Programming Exercise – Qualitative Analysis

[To be completed after running the experiments in Section 2 of the assignment.]

References

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